

MEKHMETETKA TURAN

Full-Stack Developer | React · TypeScript · Python · FastAPI · AI Integration

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PROFESSIONAL SUMMARY

Full-Stack Developer with 2 years of project and research experience building web applications, desktop software, and AI-integrated systems using React, TypeScript/Javascript, Python, and FastAPI. Sole developer of Awen, an AI language learning platform with 5 distinct exercise modes, LLM-powered evaluation, and dual TTS, released across 10 production versions. Built two research-grade tools adopted by Hof University's Visual Analytics lab. Comfortable across the full stack: REST API design, graph databases, React frontends, and CI/CD pipelines. Based in Fürth, Germany; EU Blue Card eligible on job offer.

TECHNICAL SKILLS

Frontend: React, TypeScript, Javascript, Vite, HTML5, CSS3, Tailwind CSS
Backend: Python, FastAPI, Flask, Node.js, REST API Design, Authentication / JWT
AI / ML: LangChain, Groq API, LLaMA 3, OpenCV, Whisper, NER, Relation Extraction, Prompt Engineering
Databases: PostgreSQL, SQLite, Neo4j (graph DB), SQL, ORM
DevOps: Docker, AWS, GitHub Actions, CI/CD, Linux, Git
Testing: Pytest, Vitest, Jest, Unit Testing, Integration Testing
Other: Tauri (Rust shell), Software Architecture, OOP, Agile / Scrum, REST API Design
Languages: English C2 · German B1 · Turkish C2 · Kazakh B2 · Russian B1

WORK EXPERIENCE

Student Research Assistant — Visual Analytics · Hof University of Applied Sciences Oct 2024 – Jan 2025 | Germany

Stack: Python, Flask, Neo4j, LangChain, NER, Relation Extraction

- Extended AI Spatial Hypertext post-internship**— added new NLP entity types, improved Neo4j graph traversal queries, and integrated additional LangChain processing chains, increasing recommendation accuracy for the research group.
- Collaborated directly with research supervisors**— translating academic research requirements into working software features across a tight semester schedule.

Internship — Visual Analytics · Hof University of Applied Sciences Mar 2024 – Jun 2024 | Germany

Stack: Python, Flask, Neo4j, LangChain, NER, Relation Extraction, Docker

- Architected and delivered AI Spatial Hypertext**— a human-AI collaborative knowledge tool combining a 2D spatial graph environment with an LLM chatbot; AI surfaces entity and relation recommendations, while the researcher retains full editorial control over the knowledge graph.
- Built NLP pipeline for Named Entity Recognition and Relation Extraction**— automatically populating Neo4j with structured nodes and relationships from unstructured research text, eliminating manual graph construction for the Visual Analytics group.
- Engineered LangChain-powered chat interface**— enabling natural language interaction with the Neo4j graph without writing Cypher queries; adopted and used by the Visual Analytics research group throughout the research period.
- Delivered a working research prototype as sole developer within a 3-month internship**— leading directly to a paid student assistantship extension.

Freelance Full-Stack Developer · Self-Employed Dec 2022 – Apr 2023 | Remote

Stack: React, Flask, Python, Docker, AWS, PostgreSQL

- Delivered 4 production websites for SME clients**— managing the full project lifecycle from requirements and design through deployment and post-launch maintenance.
- Built and deployed full-stack web applications across e-commerce and business landing page projects using React, Flask, Docker, and AWS**— working independently from brief to launch on each engagement.

Website Manager & Designer · Regnum TOO Jun 2022 – Nov 2022 | Almaty, Kazakhstan

Stack: WordPress, HTML, CSS, E-commerce

- Managed and optimised company website and e-commerce content**— supporting digital marketing activities and maintaining product catalogue accuracy.

Programming Teacher (Volunteer) · Community Integration Programme Apr 2025 – Jul 2025 | Nuremberg, Germany

- Taught programming fundamentals to 15 refugees across 12 weekly sessions**— adapting technical concepts for learners with no prior coding experience.

EDUCATION

B.Sc. Computer Science · Hof University of Applied Sciences 2023 – 2026 | Germany

Grade: 2.4 · Thesis: 2.3 · English · Dual-degree with SDU University, Kazakhstan (2020–23) · Thesis: Hybrid AI framework combining Generative AI with spatial hypertext for collaborative knowledge building.

KEY PROJECTS

Awen — AI-Powered German Language Learning Platform — React · TypeScript · FastAPI · Tauri · Groq LLaMA 3 · SQLite · Edge TTS

<https://github.com/etkaturan/Awen>

- Sole developer across 10 versioned releases**— 3-tier architecture: React/TypeScript frontend inside a Rust (Tauri) shell communicating with a Python FastAPI backend.
- 5 integrated exercise modes**— AI-generated listening comprehension, LLM-powered speaking partner with real-time feedback, and vocabulary flashcard builder; non-repetitive audio via LLM tokenization; real-exam quality output.
- LLM evaluation returns structured fluency, accuracy, and vocabulary scores per attempt with response times under 2 seconds**— model prompt-engineered per CEFR language level.